

Seven years of ground deformation over Vaca Muerta from free Sentinel-1 InSAR: sustained, production-correlated subsidence and an injection/extraction dipole

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Abstract

We map ground deformation across the unconventional core of the Neuquén Basin (Vaca Muerta, Argentina) using only free, public Sentinel-1 data. From the ascending track 18 archive we build a 2019–2026 Small-BASeline-Subset (SBAS) time series over a $\sim 210 \times 210$ km area with 89 acquisitions and 175 interferograms, processed with cloud-based interferograms (ASF HyP3) and MintPy, with ERA5 tropospheric correction. The line-of-sight (LOS) velocity field is stable over most of the scene (median -1.9 mm/yr; 74% of pixels pass a temporal-coherence mask) and reveals several localized subsidence bowls reaching ~ 12 mm/yr, consistent with previously published rates for the basin. The strongest-subsiding zone accumulates ~ 82 mm over seven years in a monotonic, non-seasonal trend. Across 83 concessions, mean subsidence correlates significantly with cumulative *gas* production (Spearman $\rho = -0.32$, $p = 0.004$) but not with oil or water. Two adjacent concessions display a clear injection/extraction dipole: Bandurria Norte (no active production) *uplifts* while Bandurria Sur (198 active wells) subsides, in mirror-image time series; an ascending+descending decomposition confirms both are *vertical* (Norte $+5.4$, Loma Campana -7.4 mm/yr). A Sentinel-2 NDVI cross-check does *not* support an irrigation/groundwater origin for the aggregate subsidence: strongly subsiding pixels are, if anything, slightly less vegetated than stable ground (median NDVI 0.11 vs. 0.12) over arid steppe. We frame the results as correlation, not causation: most of the area has a single line of sight and there is no local GNSS, so mechanism attribution remains open. All data and code are public.

Keywords: InSAR; Sentinel-1; land subsidence; Vaca Muerta; Neuquén Basin; unconventional hydrocarbons; induced seismicity.

1 Introduction

The Vaca Muerta formation in the Neuquén Basin is one of the largest unconventional shale plays outside North America, with intensive hydraulic fracturing, fluid injection and production since the mid-2010s. Such operations change subsurface pore pressure and can produce measurable surface deformation and induced seismicity. Brunori et al. (2022) first documented basin subsidence of -8 to -10 mm/yr with Sentinel-1 over 2017–2020 and noted a temporal association with seismicity near Sauzal Bonito, including an M_w 5 event on 7 March 2019. Schultz et al. (2024) estimate that $\sim 0.5\%$ of fracturing operations associate statistically with detectable earthquakes.

InSAR is the reference technique for millimetric subsidence (Morishita et al., 2020; Yunjun et al., 2019), and the arid/semi-arid steppe of the basin is favourable for high interferometric coherence. Here we extend the published record in three ways: (i) a longer, denser series (2019–2026) from the only locally continuous Sentinel-1 archive; (ii) a quantitative cross of subsidence against public per-concession production data; and (iii) an ascending+descending vertical decomposition resolving whether localized signals are truly vertical. We add a seismicity overlay and a Sentinel-2 NDVI test of a groundwater hypothesis. The work is a feasibility study built entirely on free data; we are explicit about what the data can and cannot establish.

2 Data and methods

SAR data and track choice. We use Sentinel-1 (Torres et al., 2012) SLC imagery on the ascending relative orbit 18, frame 1050, which has continuous coverage 2019–2026 over the study area; local descending tracks are interrupted in December 2021 after the Sentinel-1B failure. The area of interest spans -70.6 to -68.2° longitude and -39.2 to -37.3° latitude ($\sim 210 \times 210$ km) around Añelo, sampled $\sim$ monthly (89 dates).

Interferograms and time series. Interferograms were produced on demand in the cloud with ASF HyP3 (Alaska Satellite Facility, 2021) using a Small-Baseline network (each date linked to its two nearest neighbours; 175 pairs). Time-series inversion used MintPy (Yunjun et al., 2019): a coherence-based network, automatic reference-point selection, SBAS inversion, **ERA5 tropospheric correction** via PyAPS (Jolivet et al., 2011; Hersbach et al., 2020), linear deramping and DEM-error correction. We retain pixels with temporal coherence > 0.7 (4 886 991 pixels, $\sim 74\%$ of the area). The tropospheric correction is decisive: a short, uncorrected 8-month experiment produced a spurious basin-wide -17 mm/yr “subsidence” that vanished once the series was extended and corrected.

Production cross. For each concession polygon (public boundaries, Secretaría de Energía / Provincia de Neuquén) we computed the mean LOS velocity by zonal statistics and paired it with cumulative gas, oil and water production (2006–2026) from the same public source. Correlations use Pearson and Spearman coefficients on log-production.

Vertical decomposition. Combining the ascending track 18 with a descending track 112 over their spatial and temporal overlap (Añelo–Bandurria, 2019–2020), we decomposed the LOS rates into vertical and east–west components, isolating true vertical motion.

Seismicity and vegetation. We queried the ISC FDSN event service (which aggregates the Argentine INPRES reports, down to $\sim M 1.8$) for 2015–2026. For the groundwater hypothesis we built a summer-median Sentinel-2 NDVI composite (≤ 40 least-cloudy scenes, 2022–2025) from the Microsoft Planetary Computer and compared NDVI against the deformation field pixel by pixel.

3 Results

3.1 A credible regional velocity field

The LOS velocity field (Fig. 1) is stable over most of the scene (median -1.9 mm/yr; 1st/5th percentiles $-12.0/-5.3$ mm/yr; 95th/99th $+0.5/+2.1$ mm/yr). Only 2.0% of pixels subside faster

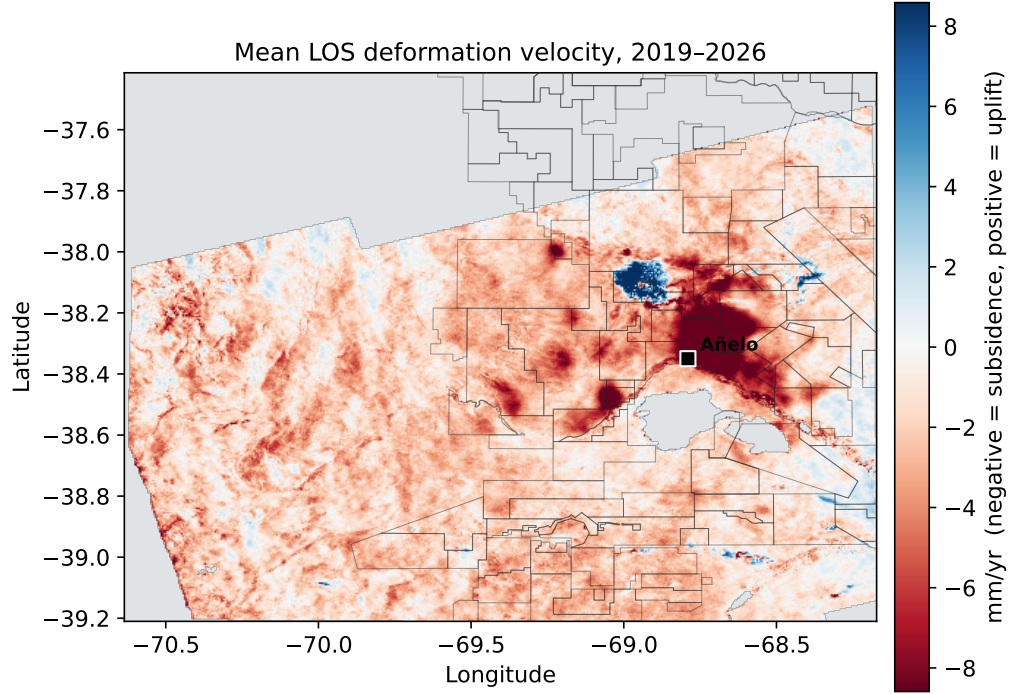


Figure 1: Mean LOS deformation velocity 2019–2026 over the study area (red = subsidence, blue = uplift). Concession outlines in grey. Stable background with localized subsidence bowls.

than -8 mm/yr, and they cluster in discrete bowls rather than as a global bias — evidence that the reference and processing are sound.

3.2 Sustained, non-seasonal subsidence

Averaging the time series over all strongly subsiding pixels (velocity < -8 mm/yr; 97 000 pixels) versus a stable reference shows a monotonic decline of ~ 82 mm over seven years (Fig. 2). The signal accumulates steadily and does not oscillate and recover, arguing against a purely seasonal artefact and for a persistent compaction process.

3.3 Subsidence correlates with gas production

Across 83 concessions with sufficient coverage, mean subsidence correlates significantly with cumulative *gas* production (Pearson $r = -0.29$, Spearman $\rho = -0.32$, $p = 0.004$; Fig. 3) but not with oil ($r = -0.18$, $p = 0.54$) or water ($r = -0.16$, $p = 0.60$). This is physically expected: extraction lowers reservoir pore pressure and the ground compacts. We stress these are observational correlations with shared confounders (geology, development era, operation type) across the shale blocks.

3.4 Bandurria: an injection/extraction dipole, confirmed vertical

The clearest single result is the contrast between two adjacent concessions (Fig. 4). Bandurria Sur (198 active wells) subsides ~ -80 mm (LOS) over seven years while Bandurria Norte (no active production) *uplifts* $\sim +80$ mm, in mirror-image time series. A block that produces nothing yet rises is the signature of fluid injection (produced-water disposal / pressure maintenance). The ascending+descending decomposition (Fig. 5) confirms the motions are vertical: Bandurria Norte

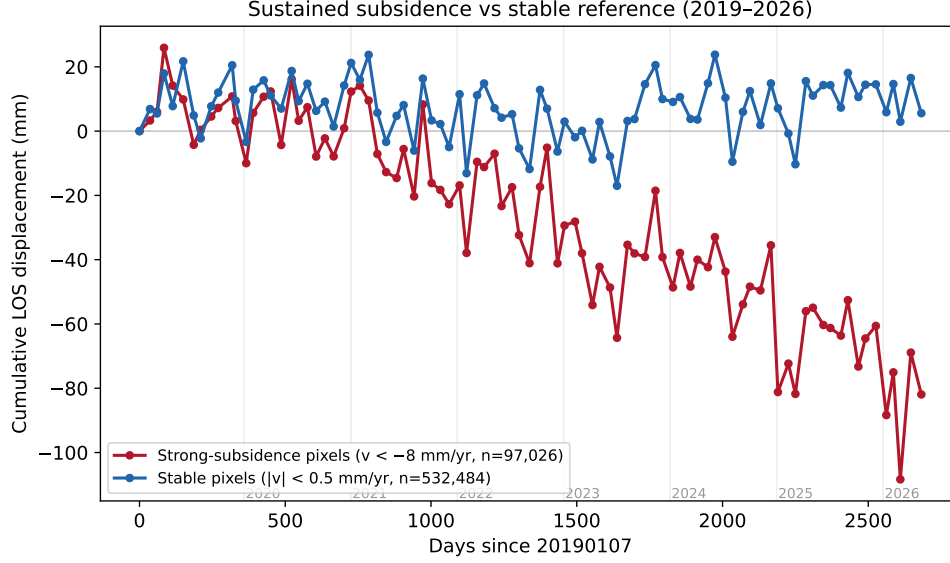


Figure 2: Cumulative LOS displacement for strongly subsiding pixels (red) versus a stable reference (blue). The subsiding aggregate spans both valley and producing blocks; it declines monotonically rather than seasonally.

+5.4 mm/yr (the only uplifting block), while producing blocks subside vertically (e.g. Loma Campana -7.4 mm/yr). The two blocks also move horizontally in opposite senses — the expected inflation/deflation pattern.

3.5 Seismicity context

The ISC/INPRES catalogue yields 70 events over the area (2015–2026), of which 67 are crustal (< 30 km, candidates for induced seismicity), concentrated to the west/south-west of Añelo around Sauzal Bonito where the literature places induced seismicity. The largest crustal event is $M 4.8$ on 7 March 2019 (reported as $M_w 5$ by Brunori et al., 2022; catalogues differ slightly for the same event). A marked swarm occurred in 2023 (41 of the 70 events). Part of the apparent increase reflects improved detection as the local network densified after ~ 2018 , so this layer is contextual rather than attributive.

3.6 The groundwater hypothesis is not supported by NDVI

Subsidence along irrigated valleys is a classic groundwater-extraction signature elsewhere (Tang et al., 2025; Ghorbani et al., 2022; Cigna and Tapete, 2021), so we tested whether the subsidence bowls coincide with irrigated vegetation. Over 8 834 069 co-located pixels the correlation between velocity and summer NDVI is weak (Spearman $\rho = -0.18$), and strongly subsiding pixels are *not* more vegetated than stable ground — median NDVI 0.11 versus 0.12 (Fig. 6), both typical of bare arid steppe (irrigated parcels reach NDVI > 0.4). At the aggregate scale, NDVI does not support an irrigation/groundwater origin; the dominant subsidence is consistent with the reservoir/production signal. A valley-restricted test against water-table records would be decisive but is currently blocked by data availability: no public piezometric time series exists for the Añelo valley (the national and provincial open repositories hold only surface-water gauging), so this cross requires institutional data (e.g. DPRH Neuquén, AIC) obtained on request, or field piezometers.

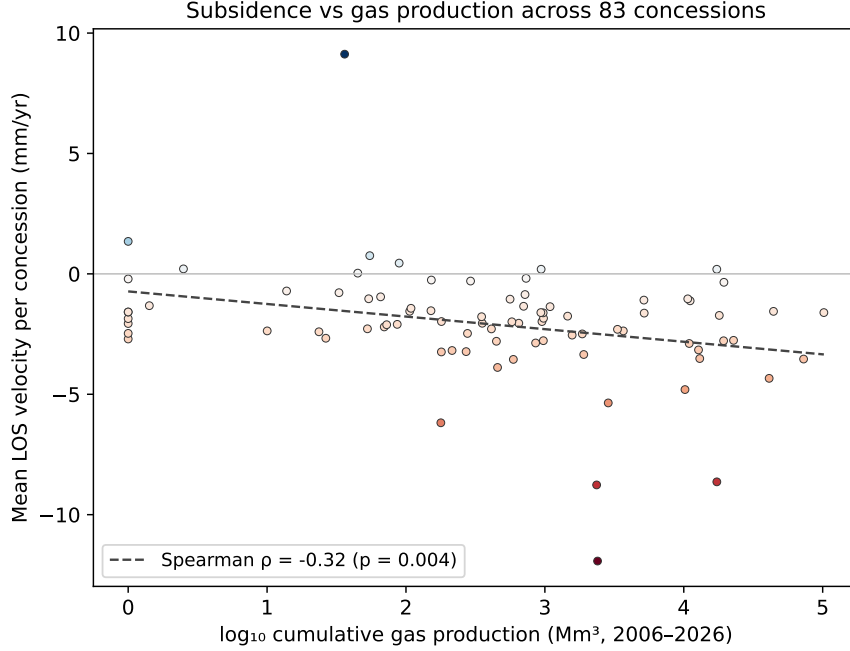


Figure 3: Mean LOS velocity per concession versus log cumulative gas production. Negative velocity is subsidence; the trend and rank correlation are significant.

4 Discussion

The data are consistent with at least two distinct subsidence signatures. The dominant one tracks hydrocarbon production: the gas–subsidence correlation, the Bandurria injection/extraction dipole, and the vertical decomposition all point to reservoir pore-pressure change (compaction under extraction, inflation under injection). A secondary, valley-located signature could in principle reflect groundwater dynamics, but the NDVI test does not support an irrigation origin at the scale examined, and the question is unresolved without water-table data.

Several caveats bound these conclusions. (i) *Correlation is not causation*; the production cross shares confounders and the seismicity association is temporal. (ii) Most of the area has a *single line of sight*; only the Bandurria region is vertically decomposed, and only for 2019–2020. (iii) There is *no local GNSS* to tie the InSAR to an absolute frame or validate rates. (iv) $\sim$ monthly sampling misses sub-monthly detail, and low-coherence zones (water, dense irrigation, active pads) carry no data.

5 Conclusions

Using only free Sentinel-1 data we obtain a credible, seven-year deformation map of the Vaca Muerta core. The background is stable; localized subsidence reaches ~ 12 mm/yr and accumulates to ~ 82 mm without seasonal recovery. Subsidence correlates with gas production, and the Bandurria injection/extraction dipole — confirmed vertical by ascending+descending decomposition — is the clearest signal of operation-driven deformation in the dataset. An NDVI cross-check does not support an irrigation-driven origin for the aggregate subsidence. The natural next steps are local GNSS, regional vertical decomposition over the full period, and a valley water-table cross to move from correlation toward attribution — the last requiring institutional piezometric records that are

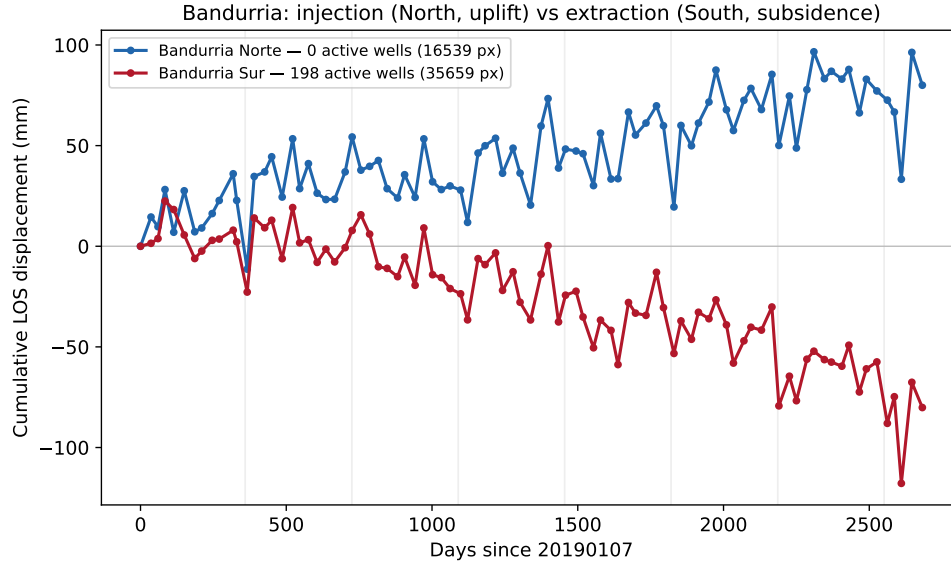


Figure 4: Mirror-image cumulative displacement: Bandurria Norte (injection, uplift) versus Bandurria Sur (extraction, subsidence).

not currently public.

Data and code availability

All processing scripts, configuration and documentation are public at <https://github.com/mpodeley/vaca-muerta-insar> with an accompanying site at <https://mpodeley.github.io/vaca-muerta-insar/>. Sentinel-1/2 data are free from ESA Copernicus / ASF and the Microsoft Planetary Computer; production and concession data are public (Secretaría de Energía / Provincia de Neuquén).

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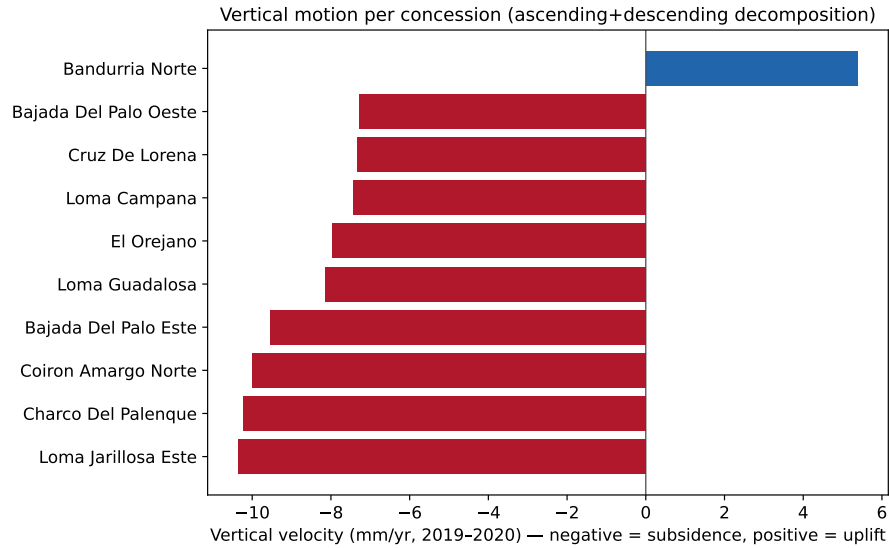


Figure 5: Vertical velocity per concession from the ascending+descending decomposition (2019–2020 overlap). Bandurria Norte is the only block that uplifts; producing blocks subside.

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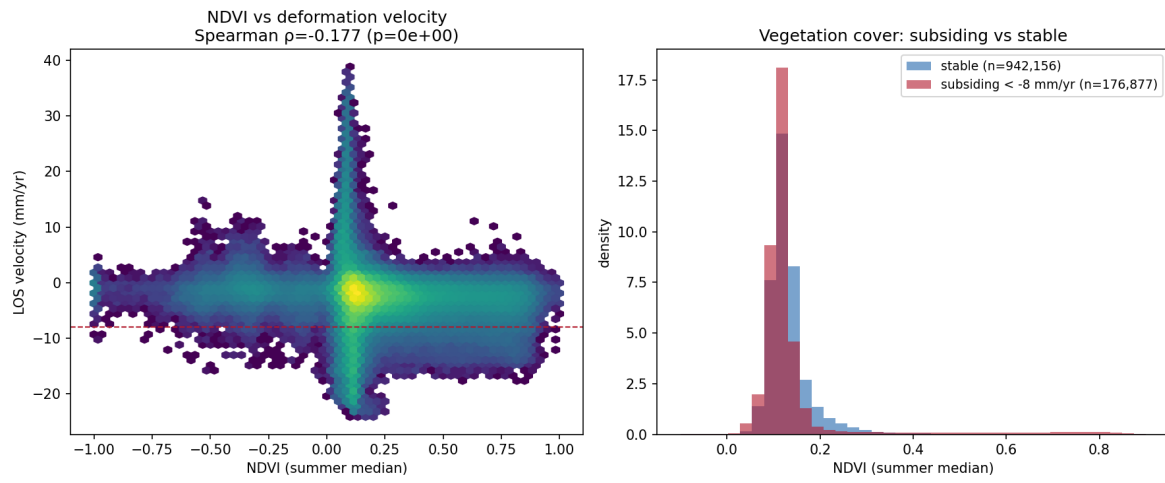


Figure 6: NDVI versus deformation velocity (left) and NDVI distribution for subsiding versus stable pixels (right). Strong subsidence does not coincide with high NDVI.